Test report 16 – Text and SVG import

Plan 17.7, step 3 (without the custom stencil font): `text()` with bundled OFL fonts, `import()` of SVG · new crates cad-outline, cad-svg, cad-font; cad-eval, cad-ir, cad-engine (color regions), cad-step (ellipses), SMLib facade (disjoint union), cad-cli, cad-desktop, web · 2026-10-23 · branch worktree-agent-a9b869d2499e391b9 · commit 6774bb4 · rustc 1.97.0 (2d8144b78 2026-07-07) · 4 cores

cargo test --workspace	clippy · fmt	Vitest · Playwright	OpenSCAD text	Wasm cad_wasm	"Hello World"
520 / 520 42 of them new/changed	**clean · clean** -D warnings	**220 · 102 / 103** new: T01–T05; C16 timing-dependent	**≤ 0.004 mm** bounding box; area ≤ 0.28 %	**+2.17 MB** 3.97 MB, gzip 1.70 MB	**505 ms** kernel native; browser 918 ms

1 Task and result

From step 3 of plan 17: **SVG import** (subset, 17.5) and **text** with bundled free fonts (17.4). The custom stencil font `"cad stencil mono"` is not part of this (it needs `stroke` from step 2); the name is reserved as the future default font; until then **JetBrains Mono** is the default font (hint for `font = "cad stencil mono"`).

- `text(t, size, font, halign, valign, spacing, direction, language, script)`: exact outlines (TrueType quadratic, CFF cubic), kerning and ligatures via `rustybuzz`, holes by nonzero, `color()` as usual. Bounding box and area against OpenSCAD 2021.01 with the same font files: bounding box ≤ 0.004 mm, area ≤ 0.28 % (OpenSCAD samples the curves).
- `import("….svg", id, scale, dpi = 96, center)`: paths (all commands), rectangles with rounding, circles and ellipses exact, polygons, groups and nested `svg` with `transform`, units, `fill-rule`, `display:none`; everything else with a warning and the SVG line.
- Files relative to the including file (CLI: file system; browser: project files; desktop: reads and writes them); changed SVGs change the cache key, `cad watch` follows them, and so does the UI scene (T02).
- Required character set with umlauts, ß/ß and punctuation in all bundled fonts (test); gaps (Allerta Stencil only) are filled from JetBrains Mono with a warning.
- Along the way: color regions of models that are only unions of their color nodes, without a Boolean with the whole (multicolor text: seconds instead of abort by the watchdog).
- Found with the alphabet chart: SMLib united disjoint glyph parts wrongly (Allerta "c": one piece twice, once reversed; volume 2.0 instead of 7.3 mm³, "a" in "abcd" disappeared, without a message). The facade now unites bodies with disjoint bounding boxes without an SMLib Boolean (`disjoint_union`: both shells into one B-rep); regression test `text_volume_is_area_times_height`.

2 Decisions

Question	Decision
Libraries	`roxmltree` 0.21 (XML, line numbers for messages), `rustybuzz` 0.20.1 with `ttf-parser` 0.25 (shaping like HarfBuzz, outlines, pure Rust, Wasm-capable). Path data, transformations, units, arcs, fill rules: own code.
Split	New crates `cad-outline` (neutral contours), `cad-svg`, `cad-font`; in `cad-eval` only the builtins (`import.rs`) and loading of data files (`Project::assets`). IR: `Seg::Nurbs` (from step 2) for B-spline chains and elliptical arcs. Engine unchanged except for the color regions; facade: union of disjoint bodies without an SMLib Boolean (below).
Profile nodes	One `Profile::Sketch` per filled SVG element or per glyph, several under a group (union): overlapping elements or glyphs give the union, as in painting; within one element its fill rule decides.
Curves	Tangent-continuous chains of Bézier curves of a glyph or a path become **one** B-spline (C1 knots, parameter lengths from the tangent lengths; exact); corners separate them. Fewer kernel faces (glyph "o": 4 instead of ~30).

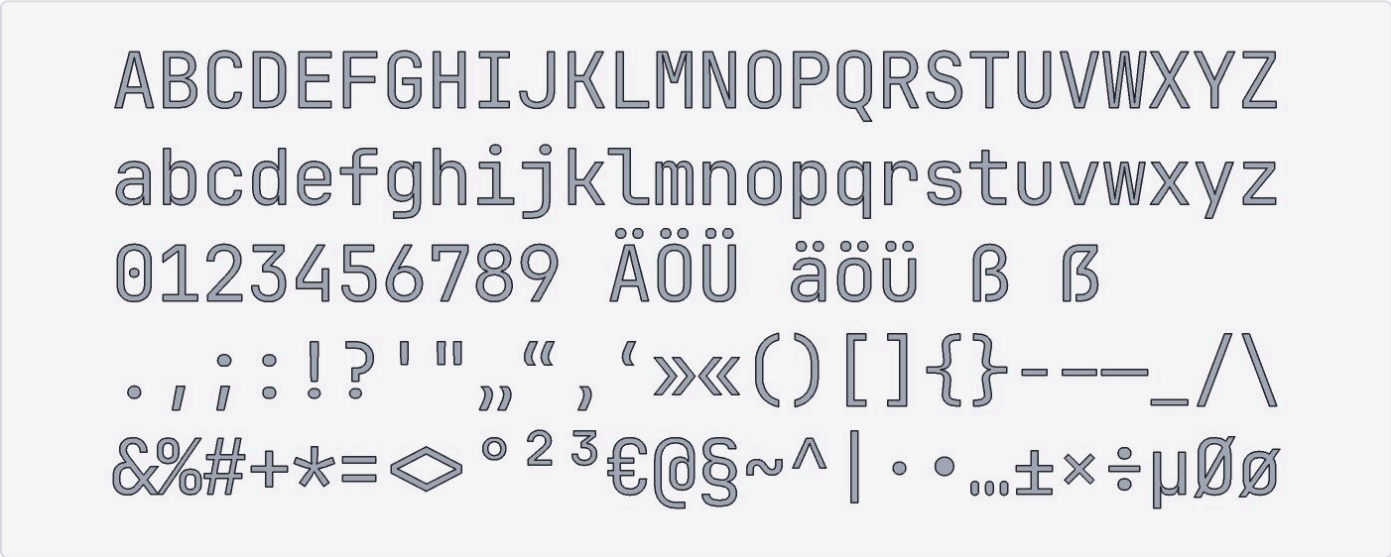
Size `size`	as in OpenSCAD: em = `size`/0.72 (OpenSCAD sets the font at `size` points at 100 dpi); cap height ≈ `size`. Advance, offset and alignment times 1000/1024 as in OpenSCAD (its FreeType conversion): text stays as wide as in OpenSCAD.
Alignment	`halign` by the advance, `valign` by the glyph bounding box (top/center/bottom), `baseline` 0; vertical text with the offsets from the shaper (centered as in OpenSCAD).
Default font	JetBrains Mono Regular until `"cad stencil mono"` exists (name reserved, hint). Unknown family: warning + JetBrains Mono; unknown style: warning + Regular of the family.
Missing characters	Per-character fallback to JetBrains Mono of the same weight (OS/2 weight ≥ 600 → Bold) with a warning at the statement: a readable character in its place is more useful than a box or a gap that is overlooked when printing; JetBrains Mono covers the required character set and much more. Allerta Stencil stays unmodified (Reserved Font Name "Allerta": modified versions may not use that name).
SVG units	as in OpenSCAD where it is consistent; default `dpi = 96` (OpenSCAD 2021.01: 72 for numbers without a unit). Deviations see 5.3.
Files	Every string in the project that ends in `.svg/.ttf/.otf/.ttc/.otc` is read on load (also parameter default values); binary files are stored in text projects (browser, `SetFiles`) as a `data:` URL. A name composed at run time that appears nowhere as a string: error "nicht geladen" ("not loaded").

3 Fonts

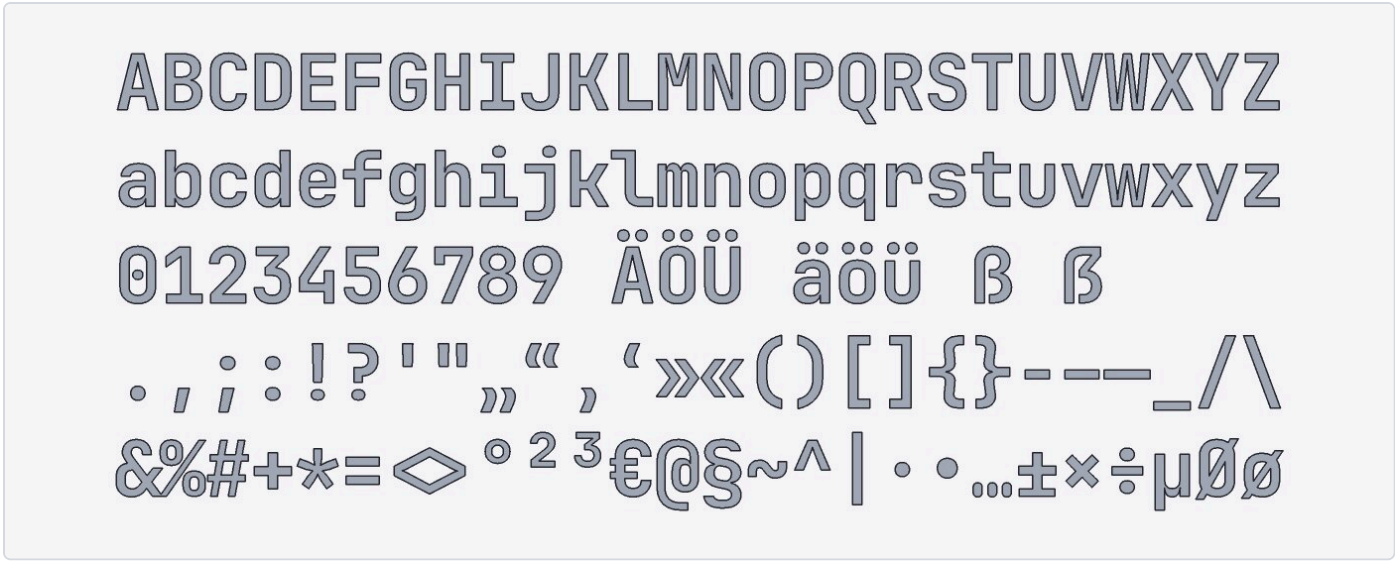
Bundled in `crates/cad-font/fonts/` (SIL OFL 1.1, license texts and sources with checksums in `fonts/README.md`), embedded with `include_bytes!`: `AllertaStencil-Regular.ttf` 20 KiB, `Inter-Bold.ttf` 411 KiB, `Inter-Regular.ttf` 402 KiB, `JetBrainsMono-Bold.ttf` 271 KiB, `JetBrainsMono-Regular.ttf` 267 KiB. Selection by family and style as in OpenSCAD (`"Inter:style=Bold"`), custom fonts by path (`font = "datei.ttf"`) or by upload in the UI (T03).

Alphabet charts

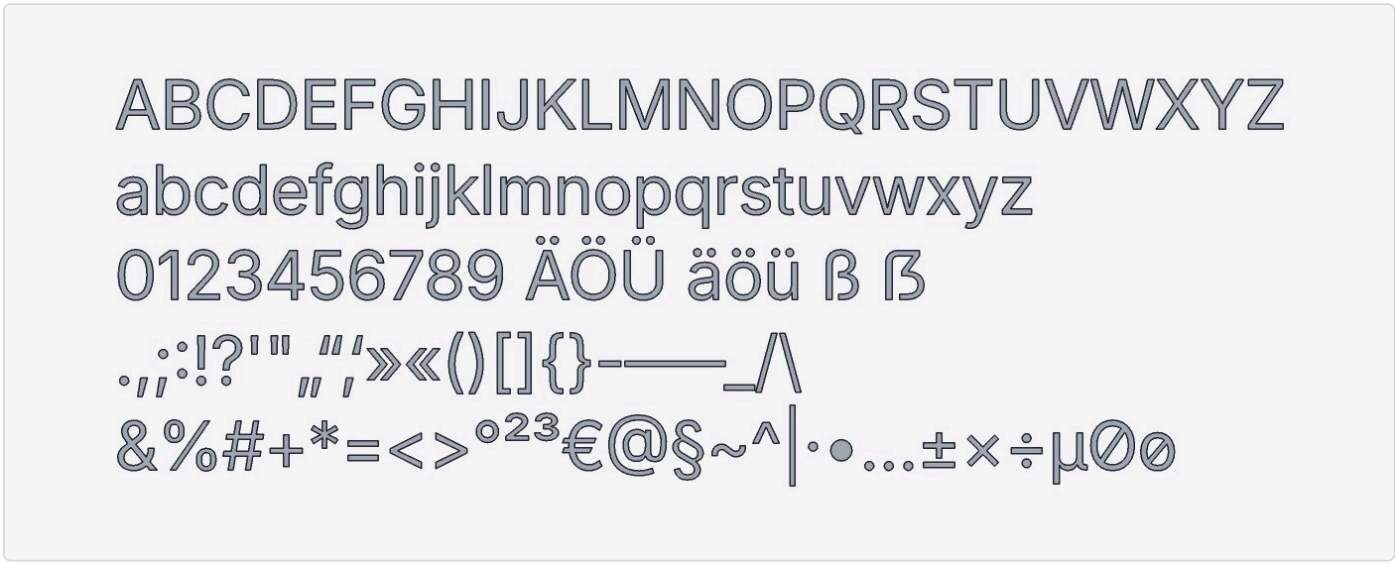
Per font `text(…, size = 8)`, extruded 0.6 mm, viewed from above (generator `tools/report/report16-data.sh`). Rows 3–5: digits, **umlauts and punctuation** of the required character set.



JetBrains Mono (default) – JetBrains Mono 2.304, Regular



JetBrains Mono:style=Bold – JetBrains Mono 2.304, Bold



Inter – Inter 4.1, Regular



Inter:style=Bold – Inter 4.1, Bold



Allerta Stencil – Allerta Stencil (Google Fonts), Regular

Required character set

`cad_font::REQUIRED_CHARS` (test `required_chars_in_every_font`: every bundled font has a real glyph for each character, no `.notdef`): A–Z a–z 0–9, Ä Ö Ü ä ö ü ß ß, space, `. , ; : ! ? ' " „ “ , ‘ » « ( ) [ ] { } - – — _ / \ & % # + * = < > ° ² ³ € @ § ~ ^ | · • …, ± × ÷ µ Ø ø`. Also applies to the future stencil font (plan 17.4).

Font	missing characters	Consequence
JetBrains Mono Regular/Bold	none	–
Inter Regular/Bold	none	–
Allerta Stencil	`ß „ , { } _ # ° ² ³ § ~ ^ · • … ± µ Ø ø`	from JetBrains Mono Regular, warning (see chart: these characters are not stenciled)

```
cases/fallback.scad:1:24: Warnung (Auswertung): Zeichen „°±“ fehlen in Allerta Stencil; ersetzt aus JetBrains Mono Regular
cases/fallback.scad: 0 Fehler, 1 Warnungen
```

4 Text against OpenSCAD 2021.01

Same font files (OpenSCAD via a fontconfig file that knows only `crates/cad-font/fonts`), `text(…)` exported as 2D (polygons) against `cad measure --bounds --volume` of a 1 mm extrusion (`tools/report/report16-openscad.py`). Bounding box in x (top) and y (bottom), mm.

Case	OpenSCAD bbox	cad bbox	max. dev.	OpenSCAD mm²	cad mm²	Area
Inter 'H'	1.221 … 9.101 0.000 … 10.105	1.221 … 9.101 0.000 … 10.105	0.0000	32.082	32.082	0.002 %
Inter:style=Bold 'H'	0.915 … 9.460 0.000 … 10.105	0.916 … 9.460 0.000 … 10.105	0.0000	49.421	49.422	0.001 %
JetBrains Mono 'H'	1.292 … 7.041 0.000 … 10.138	1.292 … 7.042 0.000 … 10.139	0.0007	29.044	29.049	0.015 %
JetBrains Mono:style=Bold 'Hamburg'	1.069 … 56.091 -2.500 … 10.138	1.069 … 56.092 -2.500 … 10.139	0.0007	235.350	235.737	0.164 %
Allerta Stencil 'H'	1.275 … 8.883 0.000 … 9.996	1.275 … 8.884 0.000 … 9.996	0.0006	37.000	37.005	0.014 %

Inter 'AV'	0.353 … 17.661 0.000 … 10.105	0.353 … 17.661 0.000 … 10.105	0.0001	54.797	54.799	0.004 %
Inter 'O'	0.827 … 9.793 -0.136 … 10.240	0.827 … 9.793 -0.136 … 10.240	0.0001	33.163	33.257	0.283 %
Inter 'Hello World'	1.221 … 73.349 -0.163 … 10.105	1.221 … 73.349 -0.163 … 10.105	0.0000	229.351	229.494	0.062 %
Inter 'Größe'	0.827 … 38.910 -0.163 … 10.430	0.827 … 38.910 -0.163 … 10.430	0.0000	130.494	130.710	0.166 %
Allerta Stencil 'ZUTRITT'	1.058 … 56.991 -0.068 … 9.996	1.058 … 56.991 -0.068 … 9.996	0.0006	185.418	185.490	0.039 %
JetBrains Mono '→'	0.986 … 15.429 0.486 … 8.680	0.985 … 15.430 0.486 … 8.681	0.0006	26.073	26.076	0.009 %
Inter 'fi'	0.136 … 7.563 0.000 … 10.579	0.136 … 7.563 0.000 … 10.579	0.0001	29.216	29.234	0.062 %
Inter 'H', halign = "center", valign = "center"	-3.819 … 4.061 -4.934 … 5.170	-3.819 … 4.061 -4.934 … 5.171	0.0004	32.082	32.082	0.002 %
Inter 'Hg', valign = "bottom"	1.221 … 17.526 -0.066 … 13.036	1.221 … 17.526 -0.070 … 13.032	0.0040	66.071	66.052	0.029 %
Inter 'Hg', valign = "top"	1.221 … 17.526 -12.866 … 0.236	1.221 … 17.526 -12.865 … 0.237	0.0009	66.071	66.052	0.029 %
Inter 'HH', spacing = 2	1.221 … 29.261 0.000 … 10.105	1.221 … 29.261 0.000 … 10.105	0.0000	64.164	64.165	0.002 %
Inter 'HH', direction = "ttb"	-3.819 … 4.061 -29.551 … -3.035	-3.819 … 4.061 -29.551 … -3.035	0.0001	64.164	64.165	0.002 %
Inter 'H', halign = "right"	-8.859 … -0.979 0.000 … 10.105	-8.859 … -0.979 0.000 … 10.105	0.0000	32.082	32.082	0.001 %
Inter 'Text', size = 25	1.662 … 68.083 -0.407 … 25.262	1.662 … 68.083 -0.407 … 25.262	0.0001	503.680	504.016	0.067 %

OpenSCAD's area is smaller because it samples the curves as polygons (chords inside the curve); the bounding box agrees to ≤ 0.001 mm (OpenSCAD writes 6 digits; its polygon vertices lie on the curve). OpenSCAD measures `valign` on rasterized glyph metrics (FreeType), hence up to 0.004 mm there. "Hello World", "AV" (kerning), "fi" and "→" (ligatures in JetBrains Mono) show that shaping and advance are identical.

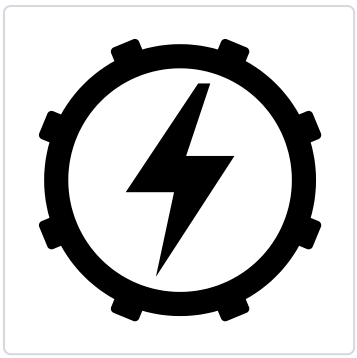
5 SVG import

5.1 Examples: source and result

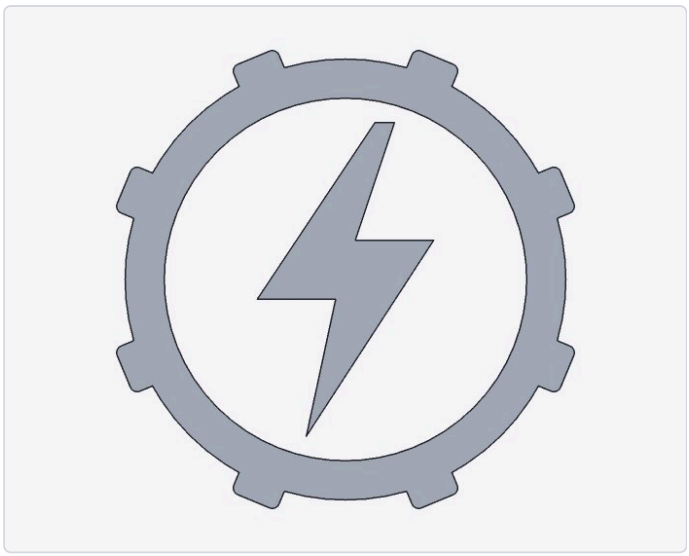
Left the SVG file as the browser draws it (source in `examples/svg/`, own drawings), right `linear_extrude(2) import("….svg")` from above.

logo.svg

Ring of half-circle arcs with a hole (evenodd), eight rotated rectangles with rounding, polygon



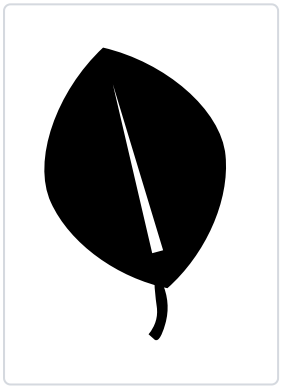
Source (browser)



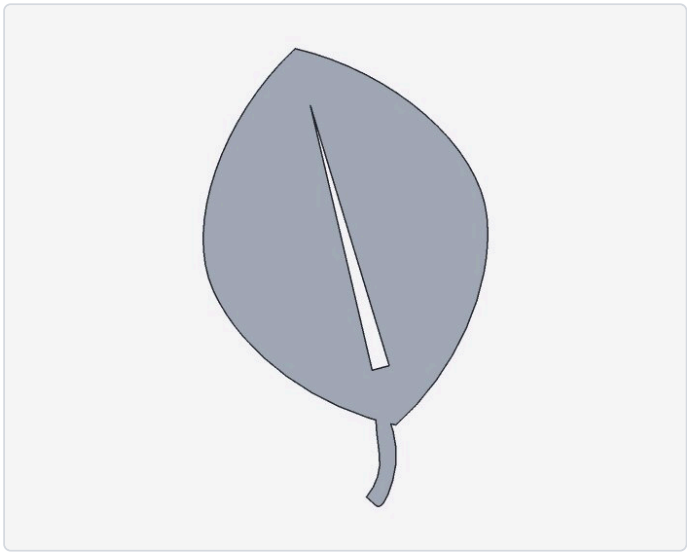
import, extruded, from above

blatt.svg

cubic Bézier (C, S), vein as reverse direction (nonzero), stem from Q/T, rotated group



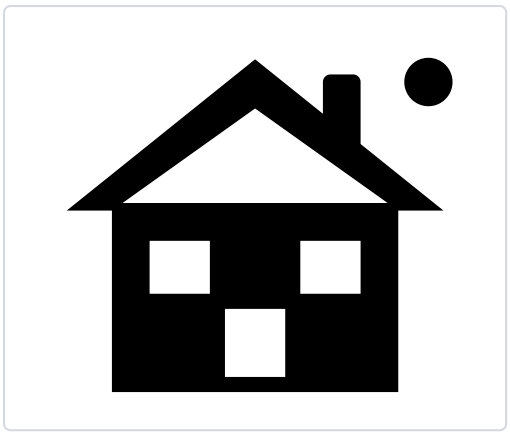
Source (browser)



import, extruded, from above

haus.svg

relative commands (h, v, m), windows and door as holes (evenodd), rectangle with rx, scaled group with a circle



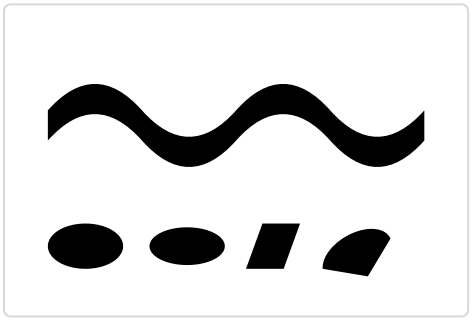
Source (browser)



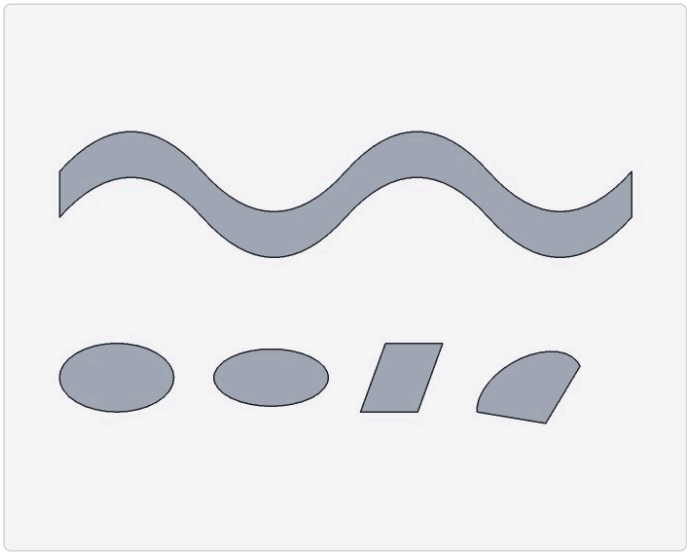
import, extruded, from above

welle.svg

Q/T wave, ellipse, circle with scale(2 1) (elliptical arcs), skewX, rotated elliptical arc (A)



Source (browser)



import, extruded, from above

5.2 Supported and unsupported

Element / property	Implementation
`path`	M L H V C S Q T A Z, absolute and relative, implicit repetition, compact numbers (`10-10.5.5`) and arc flags (`a10 10 0 0110 10`); on error the part before it is kept, warning
Arcs `A`, `circle`, `ellipse`, `rect rx/ry`	exact circular arcs (under similarity transforms), otherwise exact elliptical arcs (rational B-spline); radii too small → scaled up as in SVG
`polygon` · `polyline`, `line`, open paths	area · path only (warning "offener Pfad wird nicht gefüllt" – "open path is not filled"); paths only → open sketch (as a `sweep` path)
`g`, `a`, nested `svg`, `transform`	matrix, translate, scale, rotate (also about a point), skewX, skewY; lists from right to left
`viewBox`, `width/height`, `preserveAspectRatio`	px, mm, cm, in, pt, pc, without unit (`dpi`); origin bottom left
`fill`, `fill-rule`, `display`, `visibility`	as attribute and in `style`, inherited; `fill="none"`: not imported (warning)

Text, images, `use`, CSS classes/`<style>`, masks, clip paths, filters, gradients/patterns as fill, stroke width	warning with the SVG line (once per message); gradients are imported as a plain area
Metadata (Inkscape, Sodipodi, `defs`, `title`)	silently skipped
`id = "…"`	only this element (with its content); unknown: error

```
<svg xmlns="http://www.w3.org/2000/svg" width="40mm" height="20mm" viewBox="0 0 40 20">
  <style>.a { fill: red }</style>
  <rect class="a" width="20" height="20"/>
  <text x="22" y="12">Hallo</text>
  <image href="foto.png" width="10" height="10"/>
  <path d="M 22 2 L 38 2 L 38 8" stroke="black" fill="none"/>
  <use href="#r"/>
</svg>
```

```
cases/warn.scad:1:26: Warnung (Auswertung): warn.svg:2: CSS-Stylesheet (`<style>`) nicht unterstützt
cases/warn.scad:1:26: Warnung (Auswertung): warn.svg:3: `rect`: CSS-Klassen (`class`) nicht unterstützt; Stil
nur aus Attributen und `style`
cases/warn.scad:1:26: Warnung (Auswertung): warn.svg:4: `text`: Text nicht unterstützt (in Pfade umwandeln)
cases/warn.scad:1:26: Warnung (Auswertung): warn.svg:5: `image`: Bilder nicht unterstützt
cases/warn.scad:1:26: Warnung (Auswertung): warn.svg:6: `path`: offener Pfad wird nicht gefüllt (nur als Pfad
übernommen; Z schließt ihn)
cases/warn.scad:1:26: Warnung (Auswertung): warn.svg:6: Strichstärke (`stroke`) wird nicht als Fläche
übernommen
cases/warn.scad:1:26: Warnung (Auswertung): warn.svg:7: `use`: Verweise nicht unterstützt (Klone in Pfade
umwandeln)
cases/warn.scad:1:26: Warnung (Auswertung): warn.svg: 1 offene Pfade neben Flächen weggelassen
cases/warn.scad: 0 Fehler, 8 Warnungen
```

5.3 Units and comparison with OpenSCAD

OpenSCAD 2021.01 (`import(…, dpi = 96)`, exported as 2D) against cad. With a `viewBox` both agree. Intentional deviations (SVG rule instead of OpenSCAD quirk): **without a `viewBox`** OpenSCAD takes user units as mm but mirrors about the height in pixels (row "ohne-viewbox"); a y offset of the `viewBox` is added by OpenSCAD instead of subtracted; OpenSCAD always fills by evenodd and turns strokes into areas.

File	OpenSCAD bbox	cad bbox	max. dev.	OpenSCAD mm²	cad mm²	Area
logo.svg	1.311 … 38.689 1.311 … 38.689	1.311 … 38.689 1.311 … 38.689	0.0006	492.531	492.893	0.073 %
blatt.svg	2.039 … 26.136 2.551 … 41.319	2.033 … 26.141 2.549 … 41.319	0.0055	536.447	536.858	0.077 %
haus.svg	5.000 … 56.200 2.000 … 46.200	5.000 … 56.200 2.000 … 46.200	0.0000	1,075.692	1,075.832	0.013 %
welle.svg	2.500 … 52.500 2.000 … 27.500	2.500 … 52.500 2.000 … 27.500	0.0000	352.785	353.326	0.153 %
mm-viewbox	5.000 … 29.000 36.000 … 47.500	5.000 … 29.000 36.000 … 47.500	0.0000	100.059	100.265	0.207 %
px-viewbox	2.646 … 15.346 5.821 … 11.906	2.646 … 15.346 5.821 … 11.906	0.0000	28.018	28.076	0.207 %
in-viewbox	2.540 … 14.732 5.588 … 11.430	2.540 … 14.732 5.588 … 11.430	0.0000	25.822	25.875	0.206 %
ohne-viewbox (without viewBox)	10.000 … 58.000 -14.771 … 8.229	2.646 … 15.346 5.821 … 11.906	42.6542	400.236	28.076	92.985 %

6 Examples

examples/spec/text_schild.scad

```
// Purpose: sign with relief text (plan 17.4, report
16) — base plate with a rim, raised text in
// two fonts — multicolor (3MF with one filament per
color).
// Fonts: bundled are JetBrains Mono (default), Inter
and Allerta Stencil, each with
// a style (`"Inter:style=Bold"`); custom fonts with
`font = "datei.ttf"` (file in the project).
// `size` as in OpenSCAD: cap height about `size`.

breite = 100;
hoehe = 44;
dicke = 3;
r = 5;          // corner radius
rand = 2;       // width of the rim
relief = 1.2;   // height of text and rim

module platte(b, h, r) sketch {
    start([r, 0]);
    line(to = [b - r, 0]);
    arc(center = [b - r, r], angle = 90);
    line(to = [b, h - r]);
    arc(center = [b - r, h - r], angle = 90);
    line(to = [r, h]);
    arc(center = [r, h - r], angle = 90);
    line(to = [0, r]);
    arc(center = [r, r], angle = 90);
}

color("white") linear_extrude(dicke) platte(breite,
hoehe, r);

// rim
color("blue") translate([0, 0, dicke])
linear_extrude(relief) difference() {
    platte(breite, hoehe, r);
    translate([rand, rand]) platte(breite - 2 * rand,
hoehe - 2 * rand, r - rand);
}

// title and subtitle raised
color("red") translate([breite / 2, 26, dicke])
    linear_extrude(relief) text("Werkstatt", size =
11, font = "Inter:style=Bold", halign = "center",
valign = "center");
color("black") translate([breite / 2, 9, dicke])
    linear_extrude(relief) text("ZUTRITT NUR MIT
BRILLE", size = 4.2, font = "Allerta Stencil", halign
= "center");
```



Sign: plate from a sketch, rim, raised text in Inter Bold and Allerta Stencil, four colors



top view

```
Farben (Filament, Farbe, Volumen, Masse PLA 1,24
g/cm³):
  F1  white          #f4f3f0  13135.6195 mm³  16.29 g
Z.26
  F2  blue           #2e65b9  655.5186 mm³  0.81 g
Z.29
  F3  red            #c74240  445.7077 mm³  0.55 g
Z.35
  F4  black          #202225  117.2133 mm³  0.15 g
Z.37
```

examples/spec/svg_logo.scad

```
// Purpose: SVG import (plan 17.5, report 16) – logos
and icons from examples/svg as exact 2D profiles.
// Circles and arcs stay circular arcs, Bézier curves
stay curves, holes by fill-rule.
// Units: viewBox + width/height in mm; without
viewBox pixels with dpi (default 96).

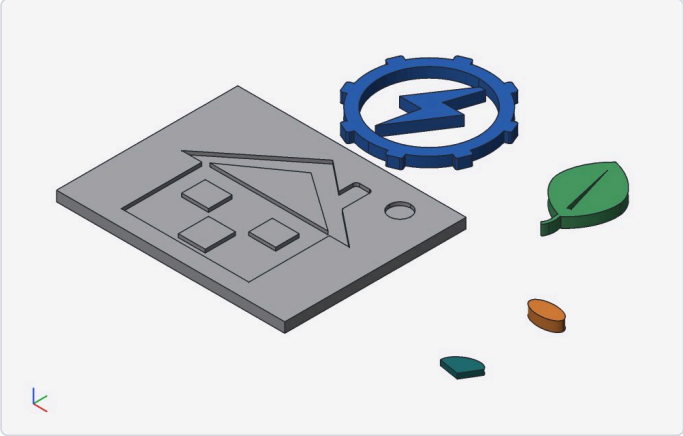
dicke = 3;

// logo, centered on the origin (center = true)
color("blue") linear_extrude(dicke)
import("../svg/logo.svg", center = true);

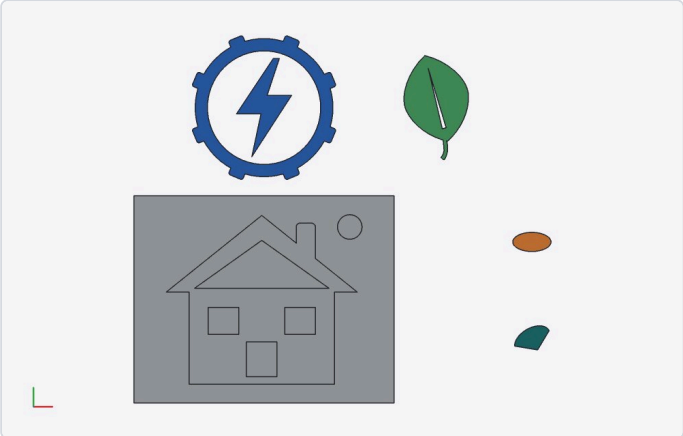
// leaf, scaled down to 70 %
translate([45, 0, 0])
    color("green") linear_extrude(dicke)
import("../svg/blatt.svg", center = true, scale =
0.7);

// house as an engraving in a plate
translate([0, -50, 0]) difference() {
    color("silver") translate([-34, -27, 0])
cube([68, 54, dicke]);
    color("red") translate([0, 0, dicke - 1])
linear_extrude(2) import("../svg/haus.svg", center =
true);
}

// only single elements of the SVG (id): non-
uniformly scaled circle (exact elliptical arcs),
// rotated elliptical arc
translate([70, -35, 0]) color("orange")
linear_extrude(dicke) import("../svg/welle.svg", id =
"oval", center = true);
translate([70, -60, 0]) color("teal")
linear_extrude(dicke / 2) import("../svg/welle.svg",
id = "bogen", center = true);
```



Logo, leaf (scaled), house as an engraving, from welle.svg only id = "oval" and id = "bogen"



top view

STEP of svg_logo: {"CIRCLE": 78, "LINE": 253, "ELLIPSE": 4, "B_SPLINE_CURVE_WITH_KNOTS": 27, "PLANE": 85, "CYLINDRICAL_SURFACE": 39, "SURFACE_OF_LINEAR_EXTRUSION": 13, "B_SPLINE_SURFACE_WITH_KNOTS": 0} – circles stay `CIRCLE`, extruded arcs `CYLINDRICAL_SURFACE`.

7 Measurements

Kernel time

Release build, `cad eval --json` without daemon, median of 3 runs (language = evaluation including shaping and SVG; kernel = all kernel calls). Browser: the same models in Chromium (SMLib as Wasm, T05).

Model	Nodes	Kernel calls	Language ms	Kernel ms native	Kernel ms browser
"Hello World" (Inter), extruded	13	20	0.7	505	918
40 characters (Inter), extruded	36	61	1.1	2,249	2,904
40 characters (JetBrains Mono)	36	61	1.2	1,917	–
"Hello World" on a plate (union)	15	24	0.7	1,449	–
examples/spec/svg_logo.scad	41	45	0.8	1,429	–
examples/spec/text_schild.scad	50	66	1.3	10,550	–

Per glyph: one extrusion per contour (outer, hole), difference for holes, one union of all glyphs (they do not touch). Text **on** a face is expensive: SMLib imprints each glyph into the top face (sign: one union of ≈ 9 s native). The new

color region rule (plan 17.10) avoids the Booleans with the whole here; before, the contact sheet of T01 took 21 s natively and failed ("Schnittmenge gescheitert" – "intersection failed"), now 4 s.

Wasm module

	raw	gzip -9
`cad_wasm_bg.wasm` before (commit cafe7a4)	1,799,439 B	702,133 B
with text/SVG (before the merge of step 2)	3,970,940 B	1,696,909 B
after the merge with sketches step 2 (state of this report)	4,120,239 B	1,757,298 B
growth from text/SVG	+2,171,501 B	+994,776 B
of which font files	1,404,056 B	–

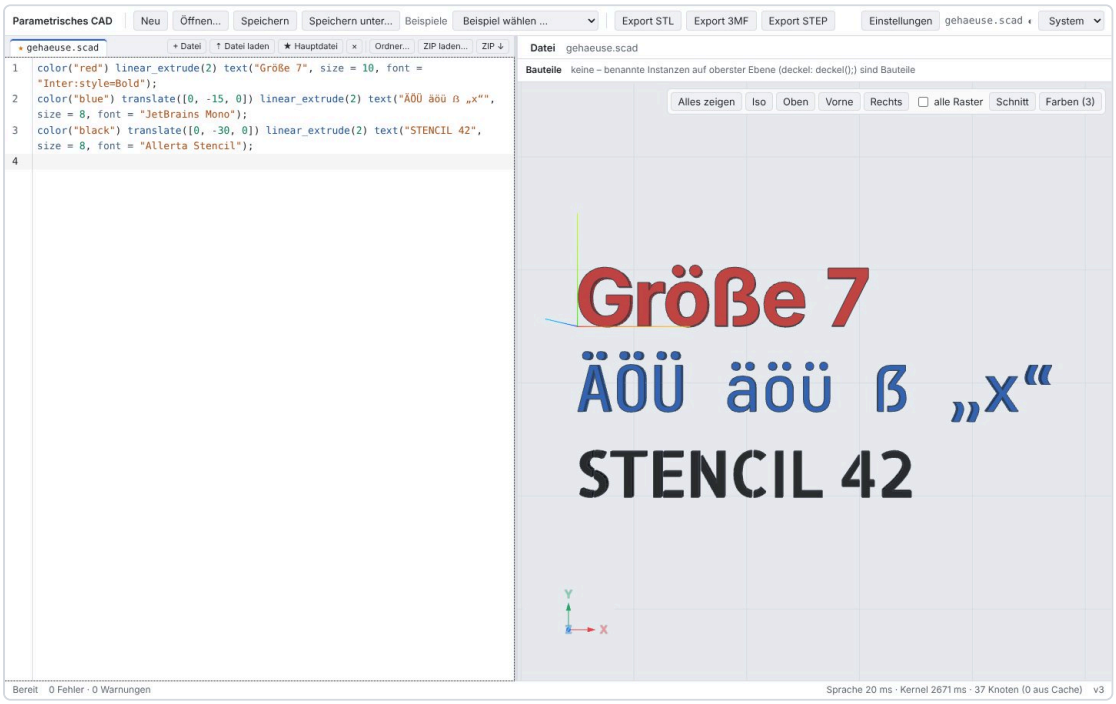
The rest of the growth is code (rustybuzz/ttf-parser, roxmltree, SVG and text logic). `smlib.wasm`: 5,285,361 B (+726 B from `disjoint_union` in the facade).

8 UI

Checks via the UI state (`window.__cad`), scenes and bounding boxes; screenshots from `web/e2e/text-svg.spec.ts`.

T01 Text in three fonts

Umlauts, ß and German quotation marks without a warning; colors as usual.



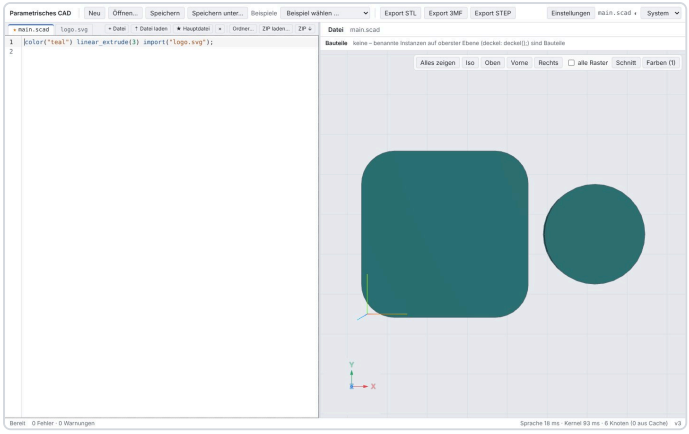
Inter Bold, JetBrains Mono, Allerta Stencil (top view)

```
color("red") linear_extrude(2) text("Größe 7", size = 10, font = "Inter:style=Bold");
color("blue") translate([0, -15, 0]) linear_extrude(2) text("ÄÖÜ äöü ß „x"", size = 8, font = "JetBrains
Mono");
color("black") translate([0, -30, 0]) linear_extrude(2) text("STENCIL 42", size = 8, font = "Allerta Stencil");
```

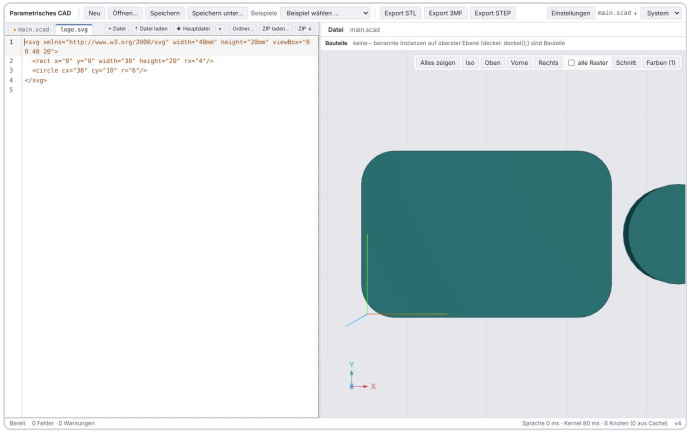
- no errors/warnings
- width > 50 mm, height 2 mm
- "20°C" in Allerta Stencil: warning, ° from JetBrains Mono

T02 SVG as a project file, changed in the editor

The SVG is a project file (tab) and is changed in the editor; the scene recomputes (cache key from the content).



Rectangle 20 mm + circle



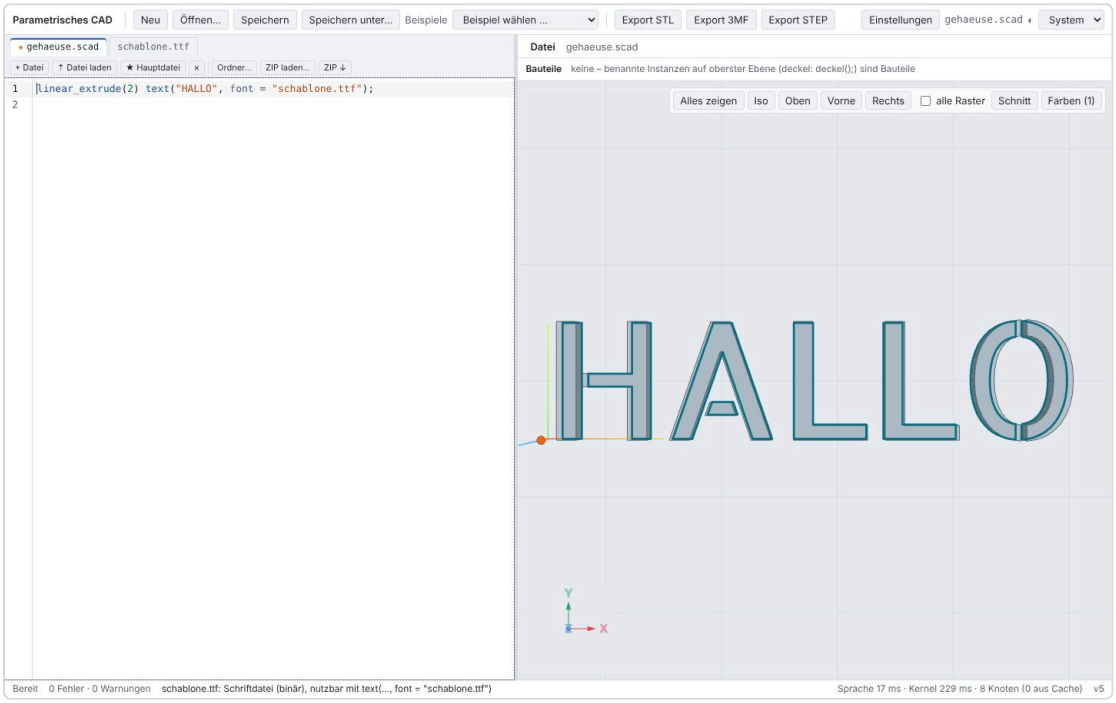
logo.svg in the editor: rectangle 30 mm

```
color("teal") linear_extrude(3) import("logo.svg");
```

- width 34 mm
- after the change 44 mm

T03 Custom font (upload)

Binary files are stored in the project as a data URL; the engine decodes them. The tab does not open an editor.

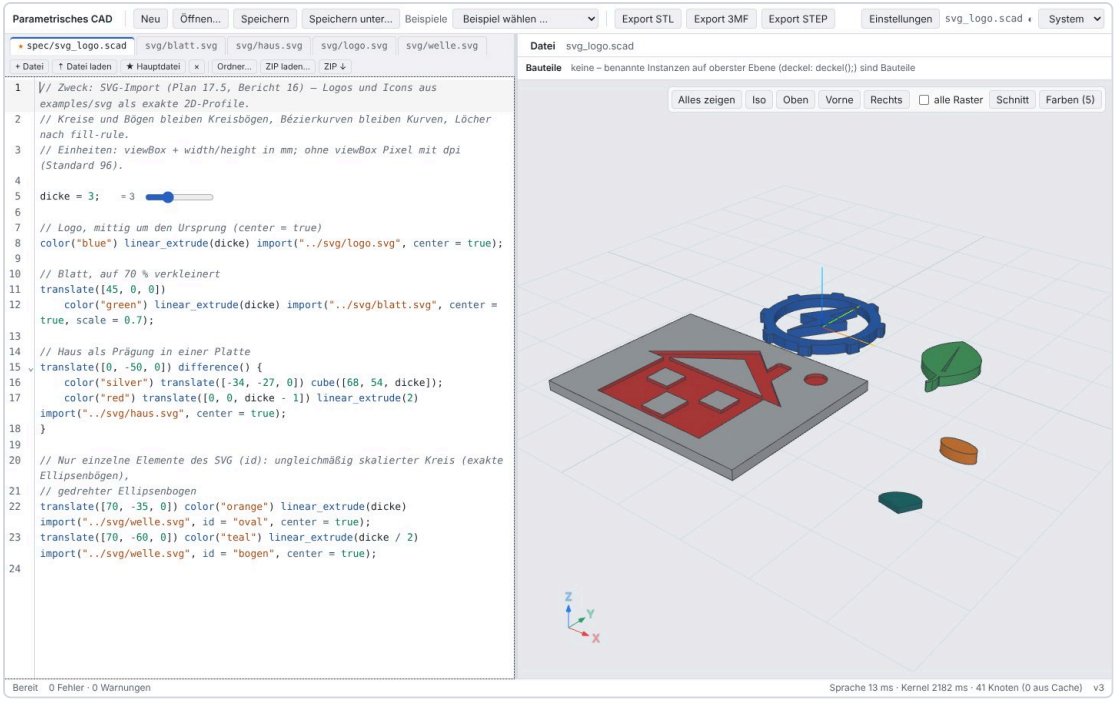


"↑ Load file" (German UI: „↑ Datei laden“): schablone.ttf, text(..., font = "schablone.ttf")

- file in the project
- no warning
- tab: hint instead of editor

T04 Example with SVG files

Single examples that import SVGs from examples/svg open as a project with these files.



examples/spec/svg_logo.scad as a project with svg/*.svg

- 4 SVG files in the project
- no errors
- <text> in the SVG: warning "w.svg:3: ...Text..."

9 Test runs

Run	Result
`cargo test --workspace --no-fail-fast`	520 passed, 0 failed
`cargo clippy --workspace --all-targets -- -D warnings` · `cargo fmt --check`	clean · clean
`pnpm typecheck` · `pnpm test` (Vitest)	clean · 220 passed
`pnpm e2e` (Playwright, Chromium)	102 / 103 – failed: C16 drag point handle constrained to an axis (Shift, X/Y/Z) · C16 (drag point handle constrained to an axis, report 3) is timing-dependent (as already in report 15): test timeout in the full run and on its own with -g C16; the whole file ui-c.spec.ts then passed 20/20 – it also failed on its own with the facade from before this step, so it is not affected by text/SVG.
`tests/openscad-compat/compare.py --eval`	36/36
`tests/step/verify.py` (OpenCASCADE)	60/60 passed
`tests/alltag/check.sh`	19 models, 0 with problems, 56 s, exports in /tmp/alltag-2sczuntc (deleted)
`node crates/kernel-smlib/js/test-node.mjs`	passed

New or changed Rust tests (42):

`import_volume_warning_and_step` **ok** · `watch_follows_a_changed_svg` **ok** · `text_volume_is_area_times_height` **ok** · `examples_check_clean` **ok** · `data_urls_roundtrip` **ok** · `import_open_paths_are_a_path` **ok** · `import_svg_gives_sketch_profiles` **ok** · `changed_svg_changes_the_tree` **ok** · `import_relative_to_the_file_and_errors` **ok** · `text_with_a_font_file` **ok** · `text_profiles_and_fonts` **ok** · `unsupported_operations_keep_children` **ok** · `tests::fallback_to_jetbrains_mono` **ok** · `tests::font_names` **ok** · `tests::glyph_o_has_a_hole` **ok** · `tests::directions` **ok** · `tests::align_and_spacing` **ok** ·

`tests::kerning_pair_av` **ok** · `tests::cff_outlines_are_cubic` **ok** · `tests::user_font_data` **ok** · `tests::hull_and_area_like_openscad` **ok** · `tests::required_chars_in_every_font` **ok** · `tests::normalize_rules` **ok** · `tests::circle_area_and_mirror` **ok** · `path::tests::arcs_exact` **ok** · `path::tests::errors` **ok** · `path::tests::commands_absolute_relative` **ok** · `path::tests::implicit_lineto_and_compact_numbers` **ok** · `path::tests::smooth_curves_reflect` **ok** · `tests::nested_svg_viewport` **ok** · `tests::hidden_and_unfilled` **ok** · `tests::errors` **ok** · `tests::open_paths_are_paths` **ok** · `tests::fill_rules` **ok** · `tests::select_by_id` **ok** · `tests::units_viewbox_like_openscad` **ok** · `tests::transforms` **ok** · `tests::shapes_exact` **ok** · `tests::units_without_viewbox_are_pixels` **ok** · `tests::unsupported_warns_with_line` **ok** · `tests::transforms_before_grid` **ok** · `spec_models_stage1` **ok**

SVG (`cad-svg`): path commands, arcs and flags, transformations, units with OpenSCAD reference values, fill rules, hidden/unfilled elements, warnings with line, `id`, error cases. Text (`cad-font`): bounding box and area against OpenSCAD (20 cases), glyph "O" with a hole, kerning "AV", alignment, directions, font names, custom font, CFF outlines, required character set, fallback. `cad-eval`: profiles, project files, errors, data URLs. CLI: examples, volume, STEP circle, `watch` follows the SVG. Vitest: data files (`web/src/logic/files.test.ts`).

10 Limitations

- The custom stencil font is missing (step 3, after `stroke`); Allerta Stencil lacks 20 characters of the required character set (taken from JetBrains Mono, not stenciled).
- Kernel time: many glyphs on a common face are expensive (the union in SMLib grows superlinearly: 9 glyphs ≈ 1.5 s, 28 ≈ 9 s native); in the browser it takes about 1.3 to 1.8 times as long; a single kernel call over 15 s triggers the watchdog.
- Fill rules assume non-intersecting contours within one element (fonts with overlapping contours, e.g. some variable fonts, and self-intersecting SVG paths are not resolved). Variable fonts only in the default instance.
- SVG: no text, no `use` clones, no CSS classes, no strokes as areas, no masks/clip paths; percent and em lengths are treated like plain numbers.
- File names must appear as strings in the project (a name composed at run time is not loaded).
- Found with the logo: the facade computed unions of extruded sketch arcs over 90° wrongly (sketches step 1); fixed in step 2 (`profile::split_arcs`), the interim workaround in the import was removed.
- STEP: the writer now writes extrusions of elliptical arcs as an extrusion of an `ELLIPSE` (OpenCASCADE had integrated the rational B-spline wrongly by $3.6 \cdot 10^{-3}$). Open: OpenCASCADE measures the extrusion of the quadratic Q/T wave from `welle.svg` 3 % too small (kernel 652.25 mm², analytical 652.19 mm²) – `svg_logo.scad` therefore takes only the ellipse and the arc from `welle.svg`; the wave itself is shown in 5.1.
- Default alignment of vertical text as in OpenSCAD (shaper offset); `valign` by the actual glyph bounding box (OpenSCAD: rasterized FreeType metrics, ≤ 0.005 mm).